

# SHIVANSH NIGAM

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## EDUCATION

Maulana Azad National Institute of Technology, Bhopal

Aug 2023 – Present

B.Tech in Electrical Engineering

**Relevant Coursework:** Data Structures and Algorithms, Computer Programming, Artificial Intelligence & Machine Learning

## SUMMARY

*Passionate problem solver with strong interests in software engineering and full-stack development. Built and deployed multiple web applications using modern technologies and solved 800+ DSA problems across coding platforms. Eager to learn, build impactful products, and contribute to real-world engineering challenges.*

## EXPERIENCE

Yasa Ed Tech

May 2026 – July 2026

Software Engineering Intern

- Enhanced existing frontend features using React.js, Next.js, TypeScript, and Tailwind CSS while maintaining a responsive and consistent user interface.
- Implemented UI improvements, integrated REST APIs, and resolved frontend issues to improve usability, performance, and overall user experience.
- Collaborated with developers using Git, participated in code reviews, and contributed to feature enhancements following Agile development practices.

## PROJECTS

SleekLink  | Node.js, Express, PostgreSQL, Redis, React, Vite, CSS3

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- Engineered a high-performance URL redirection engine using Node.js, Express, and PostgreSQL, integrating a write-through caching layer using Redis to resolve redirections with sub-millisecond latency.
- Designed an Active/Pause link control toggle with instant Redis cache invalidation, ensuring that deactivated links immediately direct visitors to a custom glassmorphic warning page (403 Forbidden) without stale-cache bypasses.
- Developed a timezone-aware visitor analytics pipeline utilizing PostgreSQL AT TIME ZONE operators to align click dates dynamically with client offsets; authored a dependency-free regex User-Agent parser to log and group device, OS, browser, and traffic referrer segments.
- Created a responsive React dashboard with zero-dependency inline SVG charts, real-time client-side table filters, and a dynamic QR Code generator utilizing virtual binary Blob downloaders to enable direct file downloads.

ShopIQ AI  | React.js, Node.js, Express.js, MongoDB, Firebase Auth, Cloudinary, Razorpay, Tailwind CSS

GitHub

- Designed and implemented a full-stack e-commerce system featuring an AI Voice-Assisted Navigator powered by the Web Speech API, enabling hands-free page navigation and vocal confirmation cues.
- Built a secure REST API in Node.js/Express with MongoDB, implementing JWT authentication, role-based admin routing gates, and image uploading pipelines with Multer.
- Developed an administrator control panel featuring a frosted-glass product editor, multi-image upload grids (synced with Cloudinary), size configuration pills, and instant status toggles.
- Integrated Razorpay payment gateway APIs alongside Cash on Delivery (COD) checkout pipelines, using the React Context API for unified, real-time shopping cart and auth state management.

Route Navigation Engine  | C++17, Node.js, Express, Redis, HTML5 Canvas, CSS

GitHub

- Engineered a high-performance pathfinding core in C++17 implementing optimized A\* Search, Dijkstra, BFS, and DFS, achieving sub-microsecond calculation latency (average 1.6µs).
- Architected a hybrid backend combining a Node.js Express API gateway with compiled C++ sub-processes, integrated with Redis caching to eliminate redundant executions and optimize query speeds.
- Designed an interactive, responsive front-end visualizer using HTML5 Canvas and CSS, featuring dynamic spatial scaling, real-time node selection, and custom path-tracing animations.

## SKILLS

**Languages:** C++, Python, JavaScript, TypeScript

**Frontend:** React.js, Next.js, HTML, CSS, Tailwind CSS

**Backend:** Node.js, Express.js, REST APIs, JWT Authentication

**Databases:** PostgreSQL, MongoDB, Redis

**Tools & Platforms:** Git, GitHub, Postman, Cloudinary, Render, Vercel

**Core CS:** Data Structures and Algorithms, Object-Oriented Programming (OOP), Database Management Systems (DBMS), Operating Systems